

PF LAB TASKS

Course Code: C1002

Subject: Programming Fundamentals

Section: C

Instructor: Mr. Shayan Ali

STUDENT SUBMISSION DETAILS

Name: AMNA KHAN

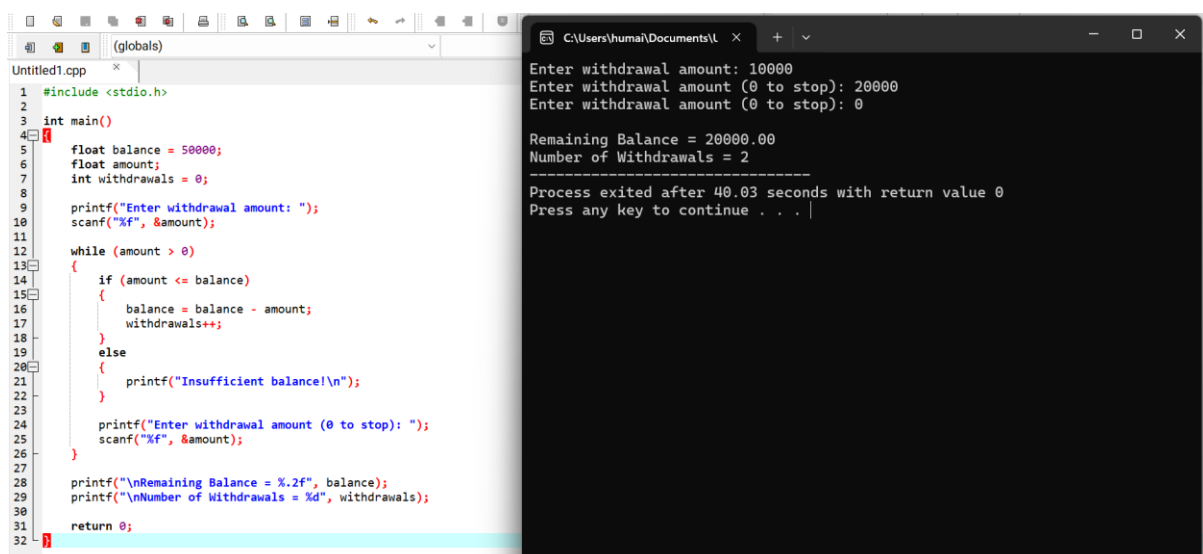
Roll No / ID: 26K-3091

Class / Section: C

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Task 01 ATM Transactions

An ATM allows a user to perform multiple withdrawal transactions. Write a C program that repeatedly asks the user to enter a withdrawal amount. Continue taking amounts while the entered amount is greater than 0. For every valid withdrawal, subtract the amount from an initial balance of 50,000. At the end, display the remaining balance and the number of withdrawals.



The image shows a C program in a code editor and its execution output in a terminal window. The code is as follows:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float balance = 50000;
6     float amount;
7     int withdrawals = 0;
8
9     printf("Enter withdrawal amount: ");
10    scanf("%f", &amount);
11
12    while (amount > 0)
13    {
14        if (amount <= balance)
15        {
16            balance = balance - amount;
17            withdrawals++;
18        }
19        else
20        {
21            printf("Insufficient balance!\n");
22        }
23
24        printf("Enter withdrawal amount (0 to stop): ");
25        scanf("%f", &amount);
26    }
27
28    printf("\nRemaining Balance = %.2f", balance);
29    printf("\nNumber of Withdrawals = %d", withdrawals);
30
31    return 0;
32 }
```

The terminal output shows the program's execution:

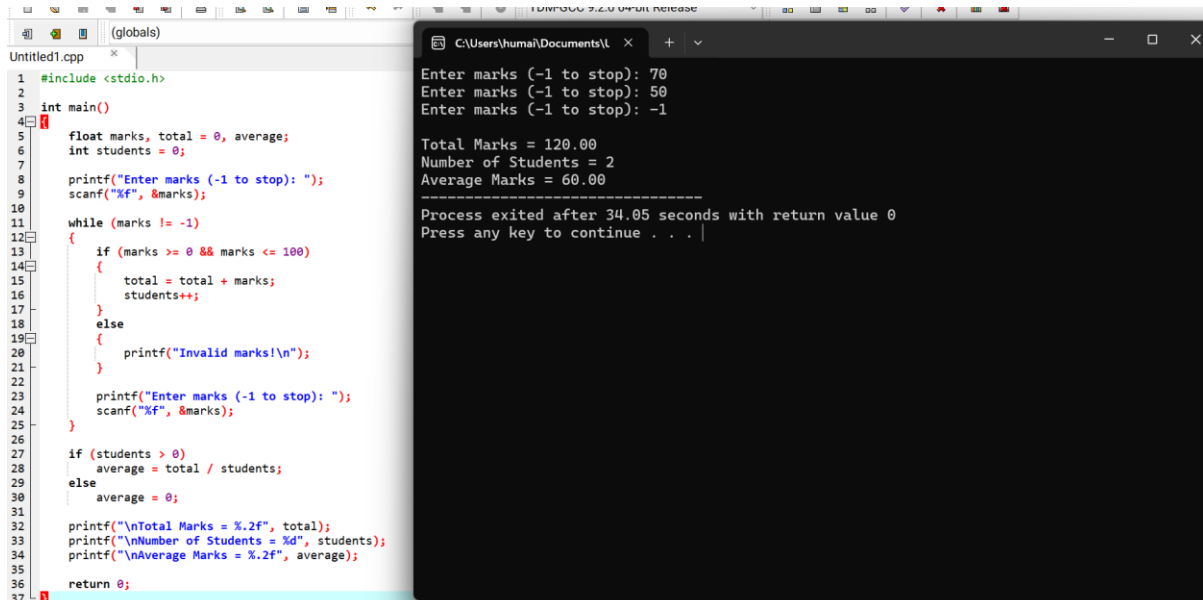
```
Enter withdrawal amount: 10000
Enter withdrawal amount (0 to stop): 20000
Enter withdrawal amount (0 to stop): 0

Remaining Balance = 20000.00
Number of Withdrawals = 2

-----
Process exited after 40.03 seconds with return value 0
Press any key to continue . . . |
```

Task 02 Student Marks

A teacher wants to calculate the total and average marks of students. Write a C program that repeatedly asks the user to enter marks between 0 and 100. Continue taking marks while the entered value is valid. When the user enters -1, stop the process and display the total marks, number of students, and average marks.



```

1 #include <stdio.h>
2
3 int main()
4 {
5     float marks, total = 0, average;
6     int students = 0;
7
8     printf("Enter marks (-1 to stop): ");
9     scanf("%f", &marks);
10
11     while (marks != -1)
12     {
13         if (marks >= 0 && marks <= 100)
14         {
15             total = total + marks;
16             students++;
17         }
18         else
19         {
20             printf("Invalid marks!\n");
21         }
22
23         printf("Enter marks (-1 to stop): ");
24         scanf("%f", &marks);
25     }
26
27     if (students > 0)
28         average = total / students;
29     else
30         average = 0;
31
32     printf("\nTotal Marks = %.2f", total);
33     printf("\nNumber of Students = %d", students);
34     printf("\nAverage Marks = %.2f", average);
35
36     return 0;
37 }

```

```

Enter marks (-1 to stop): 70
Enter marks (-1 to stop): 50
Enter marks (-1 to stop): -1

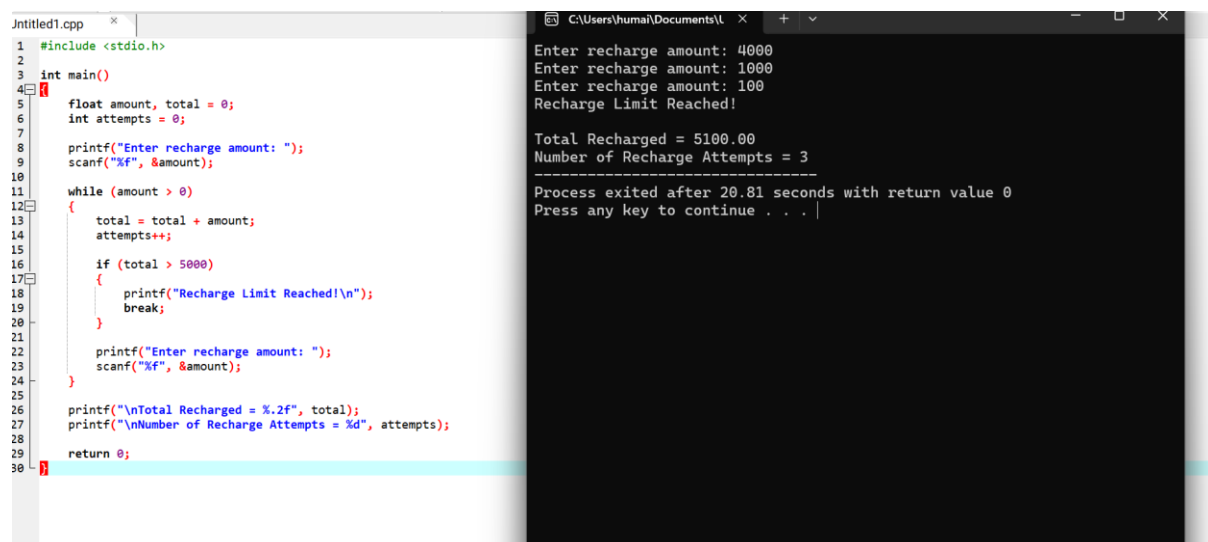
Total Marks = 120.00
Number of Students = 2
Average Marks = 60.00

-----
Process exited after 34.05 seconds with return value 0
Press any key to continue . . .

```

Task 03 Mobile Recharge

A mobile user wants to add balance several times. Write a C program that repeatedly asks for a recharge amount. Continue while the amount is greater than 0. Add every recharge amount to the total balance. If the total recharge becomes greater than 5,000, display "Recharge Limit Reached" and stop. At the end, display the total recharged amount and number of recharge attempts.



```

1 #include <stdio.h>
2
3 int main()
4 {
5     float amount, total = 0;
6     int attempts = 0;
7
8     printf("Enter recharge amount: ");
9     scanf("%f", &amount);
10
11     while (amount > 0)
12     {
13         total = total + amount;
14         attempts++;
15
16         if (total > 5000)
17         {
18             printf("Recharge Limit Reached!\n");
19             break;
20         }
21
22         printf("Enter recharge amount: ");
23         scanf("%f", &amount);
24     }
25
26     printf("\nTotal Recharged = %.2f", total);
27     printf("\nNumber of Recharge Attempts = %d", attempts);
28
29     return 0;
30 }

```

```

Enter recharge amount: 4000
Enter recharge amount: 1000
Enter recharge amount: 100
Recharge Limit Reached!

Total Recharged = 5100.00
Number of Recharge Attempts = 3

-----
Process exited after 20.81 seconds with return value 0
Press any key to continue . . .

```

Task 04 Online Shopping

You are shopping online and want to purchase multiple items. Write a C program that asks the user to enter the price of an item and then asks whether they want to add

another item (1 = Yes, 0 = No). Add the price of every item to the total. After shopping is completed, if the total is greater than 10,000, apply a 10% discount. Display the total price, discount, and final amount.

The screenshot shows a C program in a text editor and its execution in a terminal window. The program calculates the total price of items, applies a 10% discount if the total is greater than 10,000, and displays the final amount.

```

1 #include <stdio.h>
2
3 int main()
4 {
5     float price, total = 0, discount = 0, finalAmount;
6     int choice;
7
8     printf("Enter price of item: ");
9     scanf("%f", &price);
10
11     while (price > 0)
12     {
13         total = total + price;
14         printf("Do you want to add another item? (1 = Yes, 0 = No): ");
15         scanf("%d", &choice);
16
17         if (choice == 0)
18             break;
19
20         printf("Enter price of item: ");
21         scanf("%f", &price);
22     }
23
24     if (total > 10000)
25     {
26         discount = total * 0.10;
27     }
28
29     finalAmount = total - discount;
30
31     printf("\nTotal Price = %.2f", total);
32     printf("\nDiscount = %.2f", discount);
33     printf("\nFinal Amount = %.2f", finalAmount);
34
35     return 0;
36 }

```

```

C:\Users\humai\Documents\l x + v
Enter price of item: 500
Do you want to add another item? (1 = Yes, 0 = No): 1
Enter price of item: 540
Do you want to add another item? (1 = Yes, 0 = No): 0

Total Price = 1040.00
Discount = 0.00
Final Amount = 1040.00
-----
Process exited after 56.77 seconds with return value 0
Press any key to continue . . . |

```

Task 05 Restaurant Bill

A customer orders food from a restaurant. Write a C program that asks for the price of each item and whether the customer wants to order another item (1 = Yes, 0 = No). Add all item prices to calculate the total bill. If the bill is greater than 5,000, apply a 5% discount. Display the total bill, discount, and final bill.

The screenshot shows a C program in a text editor and its execution in a terminal window. The program calculates the total bill, applies a 5% discount if the total is greater than 5,000, and displays the final bill.

```

1 #include <stdio.h>
2
3 int main()
4 {
5     float price, total = 0, discount = 0, finalBill;
6     int choice;
7
8     printf("Enter price of item: ");
9     scanf("%f", &price);
10
11     while (price > 0)
12     {
13         total = total + price;
14         printf("Do you want to order another item? (1 = Yes, 0 = No): ");
15         scanf("%d", &choice);
16
17         if (choice == 0)
18             break;
19
20         printf("Enter price of item: ");
21         scanf("%f", &price);
22     }
23
24     if (total > 5000)
25     {
26         discount = total * 0.05;
27     }
28
29     finalBill = total - discount;
30
31     printf("\nTotal Bill = %.2f", total);
32     printf("\nDiscount = %.2f", discount);
33     printf("\nFinal Bill = %.2f", finalBill);
34
35     return 0;
36 }

```

```

C:\Users\humai\Documents\l x + v
Enter price of item: 550
Do you want to order another item? (1 = Yes, 0 = No): 0

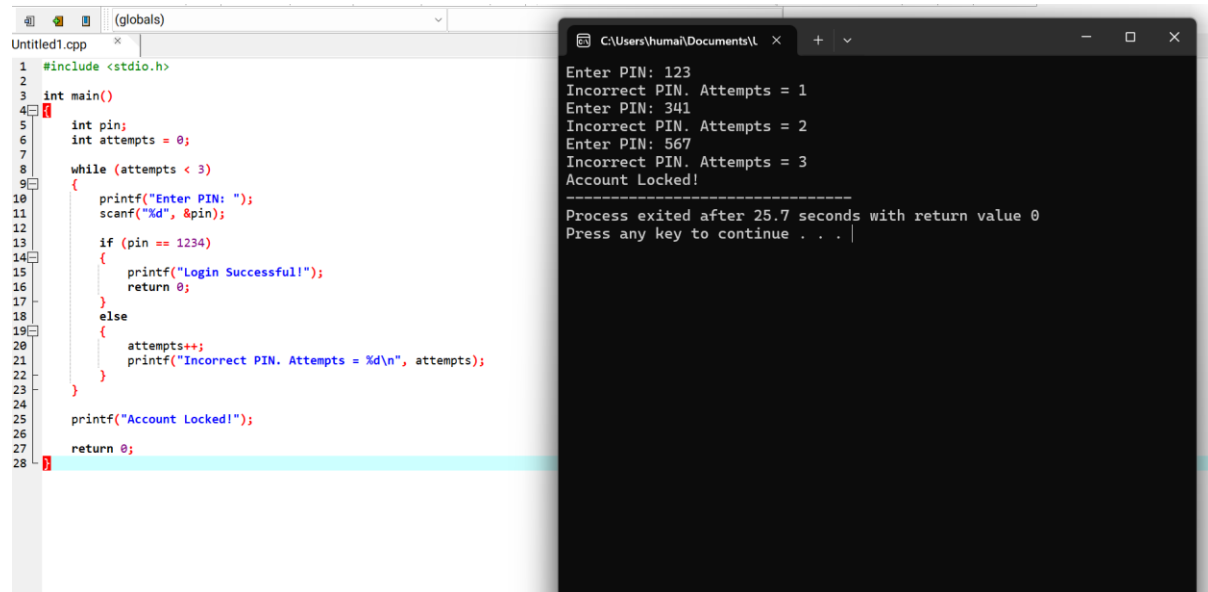
Total Bill = 550.00
Discount = 0.00
Final Bill = 550.00
-----
Process exited after 19.25 seconds with return value 0
Press any key to continue . . . |

```

Task 06 Login Attempts

A user has a maximum of 3 login attempts. Write a C program that asks the user to enter a PIN. The correct PIN is 1234. For every incorrect PIN, increase the attempt count by 1

and display the remaining attempts. If the correct PIN is entered, display "Login Successful". If all three attempts are incorrect, display "Account Locked".



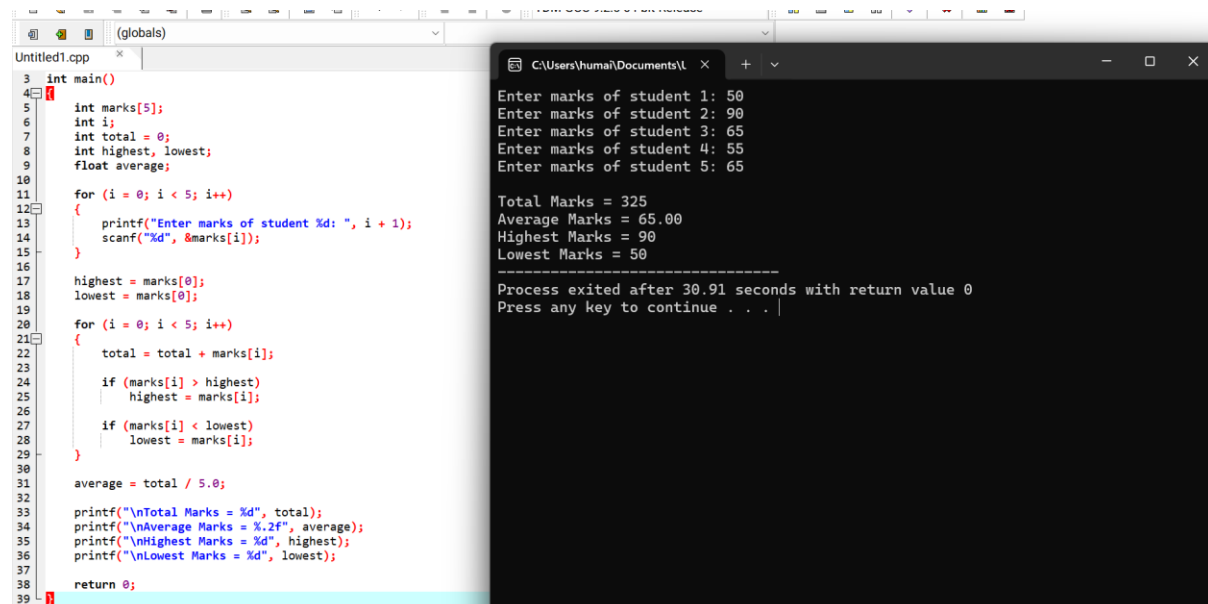
The screenshot shows a C program in a text editor and its execution output in a terminal window. The program, named 'Untitled1.cpp', includes `<stdio.h>` and defines a `main` function. It declares `int pin;` and `int attempts = 0;`. A `while` loop runs as long as `attempts < 3`. Inside the loop, it prompts the user to 'Enter PIN:', reads the input into `pin` using `scanf`, and checks if `pin == 1234`. If correct, it prints 'Login Successful!' and returns 0. If incorrect, it increments `attempts` and prints 'Incorrect PIN. Attempts = %d\n', where `%d` is replaced by the current attempt count. After the loop, it prints 'Account Locked!' and returns 0. The terminal output shows three failed attempts with PINs 123, 341, and 567, followed by 'Account Locked!' and a message indicating the process exited after 25.7 seconds.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int pin;
6     int attempts = 0;
7
8     while (attempts < 3)
9     {
10        printf("Enter PIN: ");
11        scanf("%d", &pin);
12
13        if (pin == 1234)
14        {
15            printf("Login Successful!");
16            return 0;
17        }
18        else
19        {
20            attempts++;
21            printf("Incorrect PIN. Attempts = %d\n", attempts);
22        }
23    }
24
25    printf("Account Locked!");
26
27    return 0;
28 }
```

```
Enter PIN: 123
Incorrect PIN. Attempts = 1
Enter PIN: 341
Incorrect PIN. Attempts = 2
Enter PIN: 567
Incorrect PIN. Attempts = 3
Account Locked!
-----
Process exited after 25.7 seconds with return value 0
Press any key to continue . . .
```

Task 07 Student Marks

Analysis A teacher wants to store the marks of 5 students in an array. Write a C program to take the marks and store them in the array. After taking all marks, use a loop to calculate the total marks, average marks, highest marks, and lowest marks.



The screenshot shows a C program in a text editor and its execution output in a terminal window. The program, named 'Untitled1.cpp', defines a `main` function. It declares `int marks[5];`, `int i;`, `int total = 0;`, `int highest, lowest;`, and `float average;`. A `for` loop from `i = 0` to `i < 5` prompts the user to 'Enter marks of student %d:', reads the input into `marks[i]`, and updates `total`, `highest`, and `lowest`. After the loop, it calculates `average = total / 5.0;` and prints the results: 'Total Marks = %d', 'Average Marks = %.2f', 'Highest Marks = %d', and 'Lowest Marks = %d'. The terminal output shows marks for five students (50, 90, 65, 55, 65) and the calculated statistics: Total Marks = 325, Average Marks = 65.00, Highest Marks = 90, and Lowest Marks = 50. The process exited after 30.91 seconds.

```
3 int main()
4 {
5     int marks[5];
6     int i;
7     int total = 0;
8     int highest, lowest;
9     float average;
10
11     for (i = 0; i < 5; i++)
12     {
13        printf("Enter marks of student %d: ", i + 1);
14        scanf("%d", &marks[i]);
15    }
16
17    highest = marks[0];
18    lowest = marks[0];
19
20    for (i = 0; i < 5; i++)
21    {
22        total = total + marks[i];
23
24        if (marks[i] > highest)
25            highest = marks[i];
26
27        if (marks[i] < lowest)
28            lowest = marks[i];
29    }
30
31    average = total / 5.0;
32
33    printf("\nTotal Marks = %d", total);
34    printf("\nAverage Marks = %.2f", average);
35    printf("\nHighest Marks = %d", highest);
36    printf("\nLowest Marks = %d", lowest);
37
38    return 0;
39 }
```

```
Enter marks of student 1: 50
Enter marks of student 2: 90
Enter marks of student 3: 65
Enter marks of student 4: 55
Enter marks of student 5: 65

Total Marks = 325
Average Marks = 65.00
Highest Marks = 90
Lowest Marks = 50
-----
Process exited after 30.91 seconds with return value 0
Press any key to continue . . .
```

Task 08 Shopping Cart

An online shopping system stores the prices of 5 products in an array. Write a C program to take the prices from the user and use a loop to calculate the total price. If the total is greater than 10,000, apply a 10% discount. Display the original total, discount, and final amount.

The screenshot shows a C program in a text editor and its execution in a terminal. The program, titled 'Untitled1.cpp', includes `<stdio.h>` and defines a `main` function. It declares an array `price` of size 5, a `total` variable, a `discount` variable, and a `finalAmount` variable. It uses a `for` loop to prompt the user for the price of 5 products. After the loop, it calculates the total, applies a 10% discount if the total is greater than 10000, and prints the original total, discount, and final amount. The terminal output shows the user entering prices: 50, 79, 779, 560, and 61. The program calculates an original total of 1529.00, a discount of 0.00, and a final amount of 1529.00. It also displays the execution time and a prompt to press any key to continue.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float price[5];
6     float total = 0, discount = 0, finalAmount;
7     int i;
8
9     for (i = 0; i < 5; i++)
10    {
11        printf("Enter price of product %d: ", i + 1);
12        scanf("%f", &price[i]);
13    }
14
15    for (i = 0; i < 5; i++)
16    {
17        total = total + price[i];
18    }
19
20    if (total > 10000)
21    {
22        discount = total * 0.10;
23    }
24
25    finalAmount = total - discount;
26
27    printf("\nOriginal Total = %.2f", total);
28    printf("\nDiscount = %.2f", discount);
29    printf("\nFinal Amount = %.2f", finalAmount);
30
31    return 0;
32 }
```

```
Enter price of product 1: 50
Enter price of product 2: 79
Enter price of product 3: 779
Enter price of product 4: 560
Enter price of product 5: 61

Original Total = 1529.00
Discount = 0.00
Final Amount = 1529.00

-----
Process exited after 26.48 seconds with return value 0
Press any key to continue . . . |
```

Task 09 Electricity Bill Analysis

An electricity company wants to store the electricity units consumed by 5 households in an array. Write a C program to take the units for all 5 households and use a loop to calculate the total units consumed, highest units, and lowest units. For each household, calculate the bill using 10 rupees per unit. If a household consumes more than 500 units, add an additional 5% surcharge to its bill. Finally, display the total units consumed and total amount collected.

The screenshot shows a C program in a text editor and its execution in a terminal. The program, titled 'Untitled1.cpp', includes `<stdio.h>` and defines a `main` function. It declares an array `units` of size 5, a `bill` variable, a `totalUnits` variable, and a `totalAmount` variable. It uses a `for` loop to prompt the user for the units consumed by 5 households. After the loop, it calculates the total units, the bill for each household (10 rupees per unit, plus a 5% surcharge for units over 500), and the total amount collected. It prints the total units consumed and the total amount collected. The terminal output shows the user entering units: 50, 70, 260, 240, and 780. The program calculates a total of 1400 units and a total amount of 14390.00. It also displays the execution time and a prompt to press any key to continue.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int units[5];
6     float bill, totalAmount = 0;
7     int totalUnits = 0;
8     int i;
9
10    for (i = 0; i < 5; i++)
11    {
12        printf("Enter units consumed by household %d: ", i + 1);
13        scanf("%d", &units[i]);
14    }
15
16    for (i = 0; i < 5; i++)
17    {
18        totalUnits = totalUnits + units[i];
19
20        bill = units[i] * 10;
21
22        if (units[i] > 500)
23        {
24            bill = bill + (bill * 0.05);
25        }
26
27        totalAmount = totalAmount + bill;
28    }
29
30    printf("\nTotal Units Consumed = %d", totalUnits);
31    printf("\nTotal Amount Collected = %.2f", totalAmount);
32
33    return 0;
34 }
```

```
Enter units consumed by household 1: 50
Enter units consumed by household 2: 70
Enter units consumed by household 3: 260
Enter units consumed by household 4: 240
Enter units consumed by household 5: 780

Total Units Consumed = 1400
Total Amount Collected = 14390.00

-----
Process exited after 24.32 seconds with return value 0
Press any key to continue . . . |
```